

Pulse propagation in a high-gain bichromatically-driven Raman amplifier

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Using intense electromagnetic fields, it is now possible to tailor the absorption, amplification, and dispersion properties of multi-level atoms [1]. Dramatic examples include the reduction of the group velocity of a pulse of light to 17 m/s [2] and increasing the group velocity v_g to values greater than c or even to negative values [3]. Very recently, pulses of light have been effectively stopped in an atomic medium and the information initially contained in the pulse is encoded reversibly in the atomic coherence [4].

In the recent experiment on ‘fast’ light [3] (defined as $v_g > c$ or $v_g < 0$), an alkali atom is driven by an intense bichromatic optical field that creates a gain doublet via the stimulated Raman process. Pulses of light are injected into the vapor where the spectrum of the pulse coincides with the spectral region of large anomalous dispersion between the doublet and where the size of the optical amplification is small [5]. Using a simple model, we find that the advancement of the pulse t_{adv} in comparison to its propagation time through vacuum is given by

$$\frac{t_{adv}}{\tau_p} \approx 0.13gL, \quad (1)$$

where τ_p is the pulse width, g is the line-center unsaturated gain coefficient, and L is the length of the medium. To obtain Eq. 1, we adjust the separation between the gain peaks and the bandwidth of the pulse to their optimum values. In the experiments of Wang *et al.* [3], gL was small, resulting in a small relative pulse advancement.

We describe our recent experimental observations of pulse propagation through a bichromatically driven potassium vapor in a regime where gL is large [6]. We observe several new effects that occur in this high-gain regime. One observation is that t_{adv}/τ_p is limited by a convective instability initiated by quantum fluctuations. When $gL \approx 15$ [7], quantum fluctuations at the input face of the atomic vapor cell lead to the generation of a macroscopic field that obscures the transmitted pulse and eventually depletes the pump beam power in the very high gain limit. This process of superfluorescence limits $t_{adv}/\tau_p \approx 2$.

Another observation is that the behavior of a bichromatically-driven Raman amplifier differs dramatically from its monochromatic counterpart. We observe that the bichromatic driving does more than just create a Raman gain doublet, as assumed in Ref. [3]. In addition to the Raman gain process, the bichromatic field drives four-wave mixing processes that occur even for low pulse intensities, but whose effects become most apparent when gL is large. These four-wave mixing processes lead to a dramatic broadening of the spectrum of the pulse as it propagates through the dispersion-tailored medium. The spectrum of the transmitted pulse consists of a comb of frequencies centered on pulse carrier frequency and a spacing equal to the frequency splitting of the bichromatic driving field [8]. Our observations can be motivated by an analysis of the interaction where the atom is dressed by the bichromatic field [9].

Surprisingly, we observe that the optical pulse still experiences advancement even in the presence of these complicating effects as it propagates through the high-gain amplifier. The figure shows our experimental observations for the case when $gL = 9.5$ and the splitting of the bichromatic driving field is 24 MHz. The dashed line corresponds

to the case when the pulse propagates through vacuum; it is nearly gaussian-shaped with a duration of 184 ns. The solid line shows the case when the atoms are present. It is seen that the pulse envelope is advanced and it is modulated at 24 MHz due to the creation of new frequencies via the four-wave mixing processes.

We will discuss the mechanisms responsible for the four-wave mixing and superfluorescence processes and pulse propagation through high-gain, bichromatically-driven Raman amplifiers.

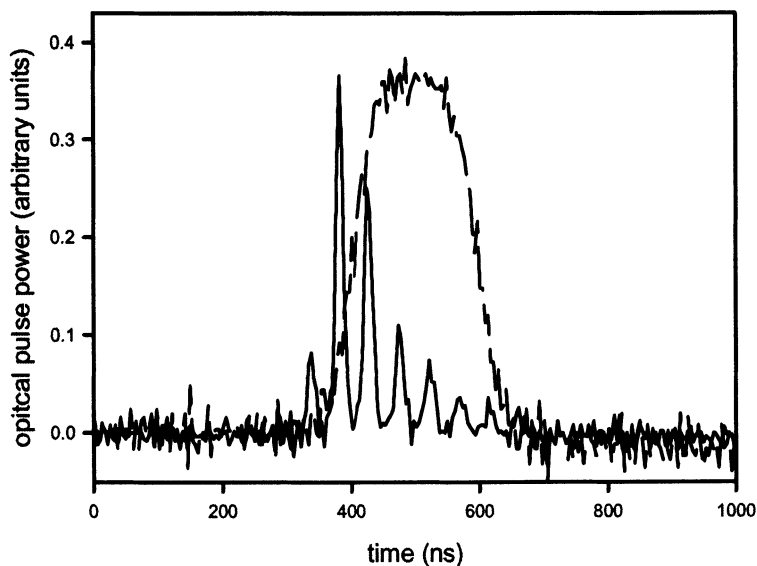


Fig. 1. Pulse propagation through vacuum (solid line) and through a high-gain bichromatically-driven Raman amplifier.

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